Problem Set 17

Partial Fractions

October 16, 2026

Problem 1

(b)

$$\int \frac{6x-2}{(x-3)(x+1)}\,dx = \int \frac{4}{x-3} + \frac{2}{x+1}\,dx$$
$$= \boxed{4\ln|x-3| + 2\ln|x+1| + C}$$

(c)

$$\int \frac{2x+5}{(x+4)(x-2)}\,dx = \int \frac{3}{2(x-2)} + \frac{1}{2(x+4)}\,dx$$
$$= \boxed{\frac{3}{2}\ln|x-2| + \frac{1}{2}\ln|x+4| + C}$$

Problem 2

(b)

$$\int \frac{x^2+4x-1}{x(x+1)(x-1)}\,dx = \int \frac{1}{x} + \frac{2}{x-1} - \frac{2}{x+1}\,dx$$
$$= \boxed{\ln|x| + 2\ln|x-1| - 2\ln|x+1| + C}$$

(c)

$$\int \frac{4x+28}{(x+1)(x-1)(x-3)}\,dx = \int \frac{3}{x+1} - \frac{8}{x-1} + \frac{5}{x-3}\,dx$$
$$= \boxed{3\ln|x+1| - 8\ln|x-1| + 5\ln|x-3| + C}$$

Problem 3

(b)

$$\int \frac{3x^2-2x-3}{x^2(x-1)}\,dx = \int \frac{5}{x} - \frac{2}{x-1} + \frac{3}{x^2}\,dx$$
$$= \boxed{5\ln|x| - 2\ln|x-1| - \frac{3}{x} + C}$$

(c)

$$\int \frac{x^2}{(x+1)^3}\,dx = \int \frac{1}{x+1} - \frac{2}{(x+1)^2} + \frac{1}{(x+1)^3}\,dx$$
$$= \boxed{\ln|x+1| + \frac{2}{x+1} - \frac{1}{2(x+1)^2} + C}$$

(d) The suggested form fails to work, as it can be simplified into a single factor. Three constants must exist, each corresponding to one of the three degrees of the denominator's polynomial.

Problem 4

(b)

$$\int \frac{x^2+x+1}{(x+1)(x^2+1)}\,dx = \int \frac{1}{2(x+1)} + \frac{x+1}{2(x^2+1)}\,dx$$
$$= \boxed{\frac{1}{2}\ln|x+1| + \frac{1}{4}\ln|x^2+1| + \frac{1}{2}\arctan x + C}$$

Problem 5

$$\int \ln(x^2-x+2)\,dx = \begin{bmatrix} u = \ln(x^2-x+2) & dv = dx \\ du = \frac{2x-1}{x^2-x+2}\,dx & v = x \end{bmatrix}$$
$$x\ln(x^2-x+2) - \int \frac{2x^2-x}{x^2-x+2}\,dx$$
$$\frac{2x^2-x}{x^2-x+2} = 2 + \frac{x-4}{x^2-x+2} = 2 + \frac{\frac{1}{2}(2x-1)-\frac{7}{2}}{x^2-x+2}$$
$$\int \ln(x^2-x+2)\,dx = x\ln(x^2-x+2) - 2x$$
$$-\frac{1}{2}\ln(x^2-x+2) + \frac{7}{2}\int \frac{1}{(x-\frac{1}{2})^2+\frac{7}{4}}\,dx$$
$$= \boxed{\left(x-\frac{1}{2}\right)\ln(x^2-x+2) - 2x + \sqrt{7}\arctan\left(\frac{2x-1}{\sqrt{7}}\right) + C}$$

x^2-x+2 is not reducible.

Problem 6

$$\int_1^2 \frac{x^2+1}{3x-x^2}\,dx = \int_1^2 -1 + \frac{3x+1}{x(3-x)}\,dx$$
$$= -1 + \int_1^2 \frac{1}{3x} + \frac{10}{3(3-x)}\,dx$$

$$= -1 + (\frac{1}{3}\ln x - \frac{10}{3}\ln(3-x))\Big|_1^2$$
$$= \boxed{-1 + \frac{11}{3}\ln 2}$$

Problem 7

(a)

$$\sin\frac{\theta}{2} = \frac{t}{\sqrt{1+t^2}},\ \cos\frac{\theta}{2} = \frac{1}{\sqrt{1+t^2}}$$
$$\sin\theta = 2\cdot\frac{t}{\sqrt{1+t^2}}\cdot\frac{1}{\sqrt{1+t^2}} = \boxed{\frac{2t}{1+t^2}}$$
$$\cos\theta = \left(\frac{1}{\sqrt{1+t^2}}\right)^2 - \left(\frac{t}{\sqrt{1+t^2}}\right)^2 = \boxed{\frac{1-t^2}{1+t^2}}$$
$$\theta = 2\arctan t \implies \boxed{d\theta = \frac{2\,dt}{1+t^2}}$$

(b)

$$\int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x} = \int_0^1 \frac{2\,dt}{t^2+2t+1}$$
$$= -\frac{2}{(t+1)}\Big|_0^1 = 2-1 = \boxed{1}$$

(c)

$$\int \frac{dx}{3\sin x - 4\cos x} = \int \frac{1}{2t^2+3t-2}\,dt$$
$$= \int \frac{2}{5(2t-1)} - \frac{1}{5(t+2)}\,dt$$
$$= \boxed{\frac{1}{5}\ln|2t-1| - \frac{1}{5}\ln|t+2| + C}$$

Problem 8

(a)

$$T = \int_{100}^{900} \frac{dP}{0.0002P(1000-P)}$$
$$= 5000\int_{100}^{900} \frac{1}{1000P} + \frac{1}{1000(1000-P)}\,dP$$
$$= 5\ln|P| - 5\ln|1000-P|\Big|_{100}^{900}$$
$$= \boxed{20\ln 3}$$

(b)

$$T_1 = T_2 = 10 \ln 3$$

The two integrals have the same value as the function is symmetric about $P = 500$.

Problem 9

(a)

$$\sum_{k=1}^{99} \frac{1}{k(k+1)} = \sum_{k=1}^{99} \frac{1}{k} - \frac{1}{k+1} = \frac{1}{1} - \frac{1}{100} = \boxed{\frac{99}{100}}$$

(b)

$$\begin{aligned}\sum_{k=1}^{n} \frac{1}{k(k+2)} &= \frac{1}{2} \sum_{k=1}^{n} \frac{1}{k} - \frac{1}{k+2} \\ &= \frac{1}{2} \left(1 + \frac{1}{2} - \frac{1}{n+1} - \frac{1}{n+2}\right) \\ &= \boxed{\frac{3}{4} - \frac{2n+3}{2(n+1)(n+2)}} \\ \lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{k(k+2)} &= \boxed{\frac{3}{4}}\end{aligned}$$

Problem 10

(a)

$$\frac{1}{Q(x)} = \sum_{i=1}^{n} \frac{A_i}{x - a_i} \iff 1 = \sum_{i=1}^{n} A_i \prod_{j\neq i} (x - a_j)$$

$$\sum_{i=1}^{n} A_i \prod_{j\neq i} (x - a_j)\Big|_{x=a_k} = A_k \prod_{j\neq k} (a_k - a_j)\Big|_{x=a_k} = \frac{A_k}{A_k} = 1$$

(b)

$$\begin{aligned}\frac{x}{Q(x)} &= \sum_{i=1}^{n} \frac{A_i x}{x - a_i} \\ \lim_{x\to\infty} \frac{x}{Q(x)} &= 0 \\ \lim_{x\to\infty} \frac{A_i x}{x - a_i} = A_i \lim_{x\to\infty} \frac{1}{1 - \frac{a_i}{x}} &= A_i = 0 \\ \boxed{\sum_{i=1}^{n} A_i = 0}\end{aligned}$$

(c)

$$\frac{x^k}{Q(x)} = \sum_{i=1}^{n} \frac{a_i^k}{(x-a_i)\prod_{j\neq i}(a_i-a_j)}$$

$$\frac{x^{k+1}}{Q(x)} = \sum_{i=1}^{n} \frac{a_i^k}{\prod_{j\neq i}(a_i-a_j)} \cdot \frac{x}{(x-a_i)}$$

$$0 \leq k \leq n-2 \implies \lim_{x\to\infty} \frac{x^{k+1}}{Q(x)} = 0$$

$$\iff \lim_{x\to\infty} \sum_{i=1}^{n} \frac{a_i^k}{\prod_{j\neq i}(a_i-a_j)} \cdot \frac{x}{(x-a_i)} = 0$$

$$k = n-1 \implies \lim_{x\to\infty} \frac{x^{k+1}}{Q(x)} = 1$$

$$\iff \lim_{x\to\infty} \sum_{i=1}^{n} \frac{a_i^k}{\prod_{j\neq i}(a_i-a_j)} \cdot \frac{x}{(x-a_i)} = 1$$

(d)

$$\sum_{i=1}^{n} \frac{a_i^k}{\prod_{j\neq i}(a_i-a_j)} = \sum_{i=1}^{n} \frac{1}{\prod_{j\neq i}(i-j)} = 0$$

$$\prod_{j\neq i}(i-j) = (i-1)!(n-i)!(-1)^{n-i}$$

$$\sum_{i=1}^{n} \frac{(-1)^{n-i}}{(i-1)!(n-i)!} = 0$$

$$\iff \sum_{m=0}^{n-1} \frac{(-1)^{n-m-1}}{m!(n-m-1)!} = 0$$

$$\iff \sum_{m=0}^{n-1} (-1)^m \frac{(n-1)!}{m!(n-m-1)!} = 0$$

$$\iff \boxed{\sum_{m=0}^{n-1} (-1)^m \binom{n-1}{m} = 0}$$

(e)

$$Q(x) = x(x-1)(x-2)(x-3)$$

$$a_1 = 0 : A_1 = -\frac{1}{6}$$

$$a_2 = 1 : A_2 = \frac{1}{2}$$

$$a_3 = 2 : A_3 = -\frac{1}{2}$$

$$a_4 = 3 : A_4 = \frac{1}{6}$$

$$\sum_{i=1}^{4} A_i = -\frac{1}{6} + \frac{1}{2} - \frac{1}{2} + \frac{1}{6} = 0$$

$$\int \frac{dx}{x(x-1)(x-2)(x-3)}$$

$$= \int \left(\frac{-1/6}{x} + \frac{1/2}{x-1} - \frac{1/2}{x-2} + \frac{1/6}{x-3} \right) dx$$

$$= \boxed{-\frac{1}{6} \ln|x| + \frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x-2| + \frac{1}{6} \ln|x-3| + C}$$